

Q.1

Show that $\frac{d}{dz} (\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1+z^2}}$

Proof: -

let $w = \operatorname{arcsinh} z$, $\frac{dw}{dz} = ?$
suppose $\rightarrow$ (a)

$$z = \sinh w$$

$$\frac{dz}{dz} = \cosh w \cdot \frac{dw}{dz}$$

$$1 = \cosh w \cdot \frac{dw}{dz}$$

$$\frac{dw}{dz} = \frac{1}{\cosh w} \rightarrow$$
 (1)

$$\therefore \cosh^2 w - \sinh^2 w = 1$$

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$$\cosh^2 w = 1 + \sinh^2 w$$

$$\sqrt{\cosh^2 w} = \sqrt{1 + \sinh^2 w}$$

$$\cosh w = \sqrt{1 + \sinh^2 w}$$

by (1)

$$\frac{dw}{dz} = \frac{1}{\sqrt{1 + \sinh^2 w}}$$

$$\text{let } z = \sinh w$$

$$z^2 = \sinh^2 w$$

$$\frac{dw}{dz} = \frac{1}{\sqrt{1 + z^2}}$$

$$\frac{d}{dz} (\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1 + z^2}}$$

Q.2

Evaluate the integral

$$\int_0^{\pi/6} e^{it} dt$$

Sol:-

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$$\int_0^{\pi/6} e^{it} dt$$

$$= \int_0^{\pi/6} (\cos 2t + i \sin 2t) dt$$

$$= \int_0^{\pi/6} \cos 2t dt + i \int_0^{\pi/6} \sin 2t dt$$

$$= \frac{\sin 2t}{2} \Big|_0^{\pi/6} + i \left(-\frac{\cos 2t}{2} \Big|_0^{\pi/6} \right)$$

$$= \frac{1}{2} \left(\sin 2\left(\frac{\pi}{6}\right) - 0 \right) + i \frac{1}{2} \left(\cos 2\left(\frac{\pi}{6}\right) - \cos 0 \right)$$

$$= \frac{1}{2} \left(\sin \frac{\pi}{3} \right) - \frac{i}{2} \left(\cos \frac{\pi}{3} - 1 \right)$$

$$= \frac{1}{2} \left(\frac{\sqrt{3}}{2} \right) - \frac{i}{2} \left(\frac{1}{2} - 1 \right)$$

$$= \frac{\sqrt{3}}{4} - \frac{i}{2} \left(\frac{1-2}{2} \right)$$

$$= \frac{\sqrt{3}}{4} - \frac{i}{2} \left(\frac{-1}{2} \right)$$

$$= \frac{\sqrt{3}}{4} + \frac{i}{4} = \frac{\sqrt{3} + i}{4} \quad \checkmark$$

Q.3

Show that $\arcsin z = -i \log(iz + (1-z^2)^{1/2})$

Sol:

$$\arcsin z = -i \log(iz + (1-z^2)^{1/2})$$

Let

$$w = \sin^{-1} z \text{ when } z = \sin w$$

$$\text{when } w = \frac{e^{iw} - e^{-iw}}{2i}$$

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put this eq. in the form

$$(e^{iw})^2 - 2iz(e^{iw}) - 1 = 0$$

~~Let~~ by solving e^{iw} we get

$$e^{iw} = iz + (1-z^2)^{1/2} \quad \text{--- (1)}$$

Here $(1-z^2)^{1/2}$ is a double-valuedfunction of z . taking $\log$ both side

of (1)

$$\log e^{iw} = \log(iz + (1-z^2)^{1/2})$$

$$iw = \log(iz + (1-z^2)^{1/2})$$

multiply with i both side

$$i^2 w = i \log(iz + (1-z^2)^{1/2})$$

$$-w = i \log(iz + (1-z^2)^{1/2})$$

$$w = -i \log(iz + (1-z^2)^{1/2})$$

$$\therefore w = \sin^{-1} z$$

$$\sin^{-1} z = -i \log(iz + (1-z^2)^{1/2})$$

$$\arcsin z = -i \log(iz + (1-z^2)^{1/2})$$

Q.4

Compute $V_p[f]$ for $f = x^2 y z$ with $p = (1, 1, 0)$ and $v = (1, 0, -3)$

Soln:

$$f = x^2 y z$$

$$\frac{\partial f}{\partial x} = 2xy z$$

$$\frac{\partial f}{\partial y} = x^2 z$$

$$\frac{\partial f}{\partial z} = x^2 y$$

at point $p(1, 1, 0)$

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0, \quad \frac{\partial f}{\partial z} = 1$$

So,

$$V_p[f] = \frac{\partial f}{\partial x} \cdot v_1 + \frac{\partial f}{\partial y} \cdot v_2 + \frac{\partial f}{\partial z} \cdot v_3$$

$$= (0)(1) + (0)(0) + (1)(-3)$$

$$= -3$$

Q.5

Write $-3i^4$ in the form $a+ib$

$$a+ib = -3i^4$$

$$= -3(i^2)^2$$

$$= -3(-1)^2$$

$$= -3+0i$$

Q.6

Show that $\log(1+i)^2 = 2\log(1+i)$
and $\log(-1+i)^2 \neq 2\log(-1+i)$

Ans:-

$$\textcircled{1} \log(1+i)^2 = 2\log(1+i)$$

Let

$$\begin{aligned} (1+i) &= \sqrt{2} \left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} \right) = \sqrt{2} \left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \right) \\ &= \sqrt{2} (e^{i\pi/4}) \end{aligned}$$

$$(1+i)^2 = (\sqrt{2} e^{i\pi/4})^2 = 2 e^{i\pi/2} = \sqrt{2} e^{i\pi/4}$$

$$\text{therefore Arg}(1+i) = \frac{\pi}{4} \text{ and Arg}(1+i)^2 = \frac{\pi}{2}$$

$$\begin{aligned} \log(1+i)^2 &= \ln|(1+i)^2| + i\text{Arg}(1+i)^2 \\ &= \ln|2e^{i\pi/2}| + i\frac{\pi}{2} \\ &= \ln|2| + i\frac{\pi}{2} \\ &= 2 \left(\frac{1}{2} \ln|2| + i\frac{\pi}{4} \right) \\ &= 2 (\ln\sqrt{2} + i\frac{\pi}{4}) \\ &= 2\log(1+i) \end{aligned}$$

Q.6-Part 2

$$\log(-1+i)^2 \neq 2\log(-1+i)$$

L.H.S

$$-1+i = \sqrt{2}\left(-\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right) = \sqrt{2}e^{i\frac{3\pi}{4}}$$

$$(-1+i)^2 = (\sqrt{2}e^{i\frac{3\pi}{4}})^2 = 2e^{i\frac{3\pi}{2}}$$

$$\text{therefore } \text{Arg}(-1+i) = \frac{3\pi}{4} \text{ and } \text{Arg}(-1+i)^2 = \frac{3\pi}{2}$$

$$\log(-1+i)^2 = \ln|(-1+i)^2| + i\text{Arg}(-1+i)^2$$

$$= \ln|2e^{i\frac{3\pi}{2}}| + i\frac{3\pi}{2}$$

$$= \ln|2|(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}) + i\frac{3\pi}{2}$$

$$= \ln|2| + i\frac{3\pi}{2} \neq 2\log(-1+i)$$

thus,

$$\log(-1+i)^2 \neq 2\log(-1+i)$$